It is comprised of 3 main elements: the Nosecone, which protects the vessel and the docking adaptor during ascent; the Spacecraft, which houses the crew and/or pressurized cargo as well as the service section containing avionics, the RCS system, parachutes, and other support infrastructure; and the Trunk, which provides for the stowage of unpressurized cargo and will support Dragon's solar arrays and thermal radiators.

To ensure a rapid transition from cargo to crew capability, the cargo and crew configurations of Dragon are almost identical, with the exception of the crew escape system, the life support system and onboard controls that allow the crew to take over control from the flight computer when needed. This focus on commonality minimizes the design effort and simplifies the human rating process, allowing systems critical to Dragon crew safety and ISS safety to be fully tested on unmanned demonstration flights.

For cargo launches the inside of the spacecraft is outfitted with a modular cargo rack system designed to accommodate pressurized cargo in standard sizes and form factors. For crewed launches, the interior is outfitted with crew couches, controls with manual override capability and upgraded life-support.

With the Space Shuttle retired, NASA has announced the selection of SpaceX's Falcon 9 launch vehicle and Dragon spacecraft to resupply the International Space Station. The first mission is scheduled for February 2012. Though designed to address cargo and crew requirements for the ISS, the free-flying spacecraft Dragon also provides an excellent platform for in-space technology demonstrations and scientific instrument testing. SpaceX is also currently manifesting fully commercial, non-ISS Dragon flights under the name "DragonLab". DragonLab represents an emergent capability for in-space experimentation.

Commercial Space Station

While Virgin Galactic and SpaceX are the two firms fighting to be the first to commercial spaceflight, there are two major contenders in the fight to launch the world's first commercial space station. But in the case of both Bigelow Aerospace and Orbital Technologies, it's still a dream quite far away with the technology going unproven so far, apart from a couple of satellites launched by Bigelow.

- **Orbital Technologies' Commercial Space Station –** The Commercial Space Station (CSS) will be man-tended, with a crew capability of up to seven people. Space-certified elements, modules and technologies of the highest quality and reliability will be used in the construction of the station. The CSS will be serviced by the Russian Soyuz and Progress spacecraft, as well other transportation systems available in the global